

# PS3: ARCH, GARCH

Michael Fehl, Francesco Zucca, Giuseppe Lista

2026-03-08

## Table of contents

|                                          |          |
|------------------------------------------|----------|
| <b>1 Introduction</b>                    | <b>2</b> |
| <b>2 Questions</b>                       | <b>2</b> |
| 2.1 Download Data . . . . .              | 2        |
| 2.2 Graphical Representation . . . . .   | 3        |
| 2.3 Compute Log>Returns . . . . .        | 4        |
| 2.4 Model Fitting . . . . .              | 5        |
| 2.4.1 ARCH(1) . . . . .                  | 5        |
| 2.4.2 ARCH(3) . . . . .                  | 6        |
| 2.4.3 GARCH(1,1) . . . . .               | 7        |
| 2.5 Fitted Model Expressions . . . . .   | 7        |
| 2.5.1 ARCH(1) . . . . .                  | 8        |
| 2.5.2 ARCH(3) . . . . .                  | 9        |
| 2.5.3 GARCH(1,1) . . . . .               | 10       |
| 2.6 Results & Best Fit . . . . .         | 12       |
| 2.6.1 Information Criteria . . . . .     | 12       |
| 2.6.2 Coefficient Significance . . . . . | 13       |
| 2.6.3 Ljung-Box Test . . . . .           | 14       |
| 2.6.4 Jarque-Beta/Shapiro-Wilk . . . . . | 15       |
| 2.6.5 LM Arch Test . . . . .             | 16       |
| 2.6.6 Model Decision . . . . .           | 16       |

```
library(tidyverse)  # tidy data ecosystem
library(patchwork)  # patch ggplots
library(fGarch)     # ARCH, GARCH modeling
library(fBasics)    # basicStats()
```

# 1 Introduction

In this practice, we will work on fitting different volatility models (ARCH and GARCH). We will also focus on discussing the validation and selection of the model. As done in previous practices, each student group must follow the practice script below, performing the necessary calculations and graphs in the R environment. The obtained results and answers to the questions should be compiled into a document containing your solution, created, for example, in R Markdown or Microsoft Word. Please make sure to include your full names and group number at the beginning of the document. Submit the questionnaire solution in PDF format by uploading it to the platform under the corresponding task section before the deadline.

In this questionnaire, we will work with the following time series:

- KO: The Coca-Cola Company (KO) stock (NYSE)

You can download the data using this code:

```
library(quantmod) # quant finc modeling and trading framework
getSymbols("KO")
```

```
[1] "KO"
```

## 2 Questions

### 2.1 Download Data

**Task:** Download the time series. You may work with the complete dataset (from 2007 to the most recent trading session).

**Solution**

```
cat('column names:', names(KO), '\n\n') # columns
```

```
column names: KO.Open KO.High KO.Low KO.Close KO.Volume KO.Adjusted
```

```
summary(KO) # quick summary stats
```

| Index              | KO.Open           | KO.High       | KO.Low        |
|--------------------|-------------------|---------------|---------------|
| Min. :2007-01-03   | Min. :19.08       | Min. :19.61   | Min. :18.72   |
| 1st Qu.:2011-10-13 | 1st Qu.:33.97     | 1st Qu.:34.21 | 1st Qu.:33.77 |
| Median :2016-08-01 | Median :42.82     | Median :43.20 | Median :42.53 |
| Mean :2016-08-01   | Mean :44.55       | Mean :44.86   | Mean :44.23   |
| 3rd Qu.:2021-05-17 | 3rd Qu.:54.60     | 3rd Qu.:54.91 | 3rd Qu.:54.26 |
| Max. :2026-03-06   | Max. :81.25       | Max. :82.00   | Max. :80.82   |
| KO.Close           | KO.Volume         | KO.Adjusted   |               |
| Min. :18.92        | Min. : 2996300    | Min. :11.18   |               |
| 1st Qu.:34.00      | 1st Qu.: 11417600 | 1st Qu.:21.87 |               |
| Median :42.92      | Median : 14403150 | Median :31.81 |               |
| Mean :44.55        | Mean : 16210680   | Mean :35.44   |               |
| 3rd Qu.:54.61      | 3rd Qu.: 18930075 | 3rd Qu.:47.46 |               |
| Max. :81.56        | Max. :124169000   | Max. :81.56   |               |

```
chartSeries(KO)
```

```
# financial chart
```

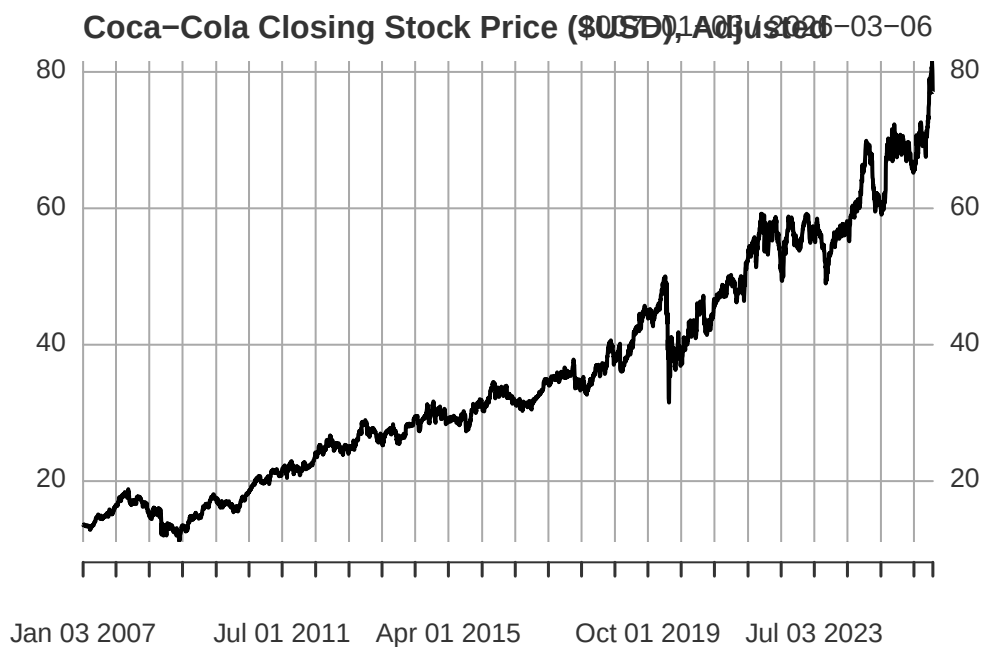


## 2.2 Graphical Representation

**Task:** Create a graphical representation of the time series using the Adjusted Closing Price.

**Solution**

```
plot(KO$KO.Adjusted,  
      main = "Coca-Cola Closing Stock Price ($USD), Adjusted")
```



## 2.3 Compute Log>Returns

**Task:** Compute the log-returns of the series. It is not necessary to model the mean (e.g., using an ARIMA model).

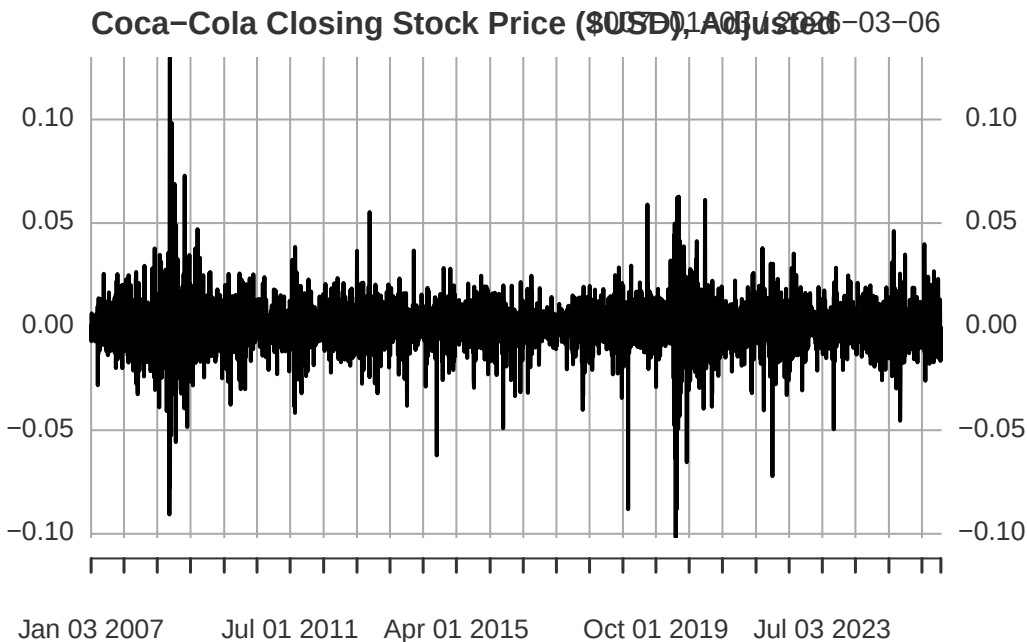
**Solution**

```
# returns: can do this directly using fQuant library
KO$log_returns <- dailyReturn(KO$KO.Adjusted,
                              type = "log")

# sanity check
head(KO$log_returns)

      log_returns
2007-01-03 0.0000000000
2007-01-04 0.0004118395
2007-01-05 -0.0070204185
2007-01-08 0.0064024905
2007-01-09 0.0008230482
2007-01-10 0.0014397037

# plot
plot(KO$log_returns,
     main = "Coca-Cola Closing Stock Price ($USD), Adjusted") # looks stationary -- good
```



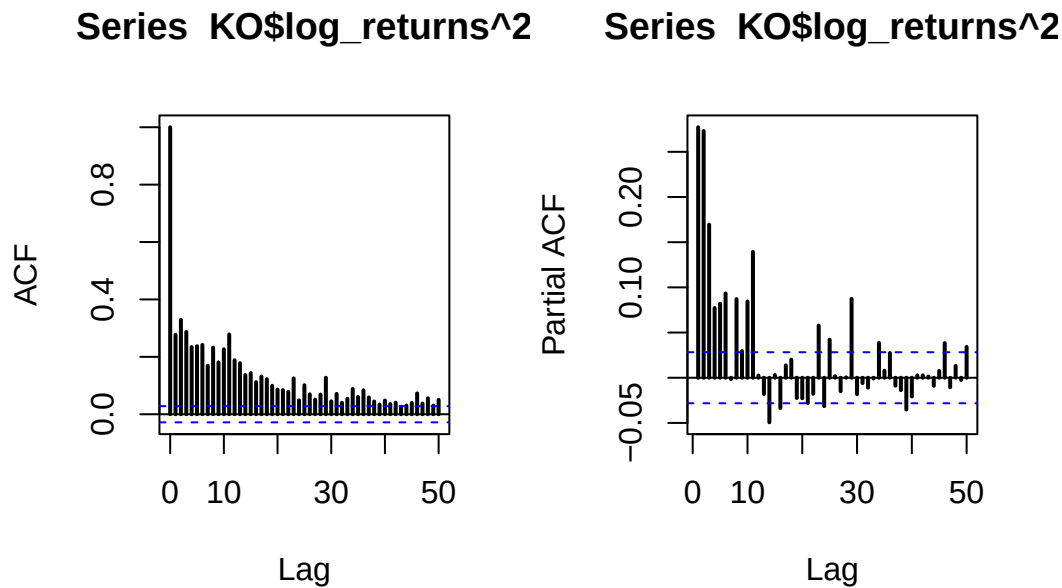
Before moving on to model fitting, we do some basic analysis of the returns.

```
stats <- basicStats(KO$log_returns)
stats[c("Skewness", "Kurtosis"), , drop = FALSE]
```

```
log_returns
Skewness  -0.160944
Kurtosis   11.303078
```

The time series is more skewed compared to the normal distribution, and it has fatter tails because of the excess kurtosis of around 11.

```
par(mfrow = c(1, 2))
acf(KO$log_returns^2, lwd = 2, lag.max = 50)
pacf(KO$log_returns^2, lwd = 2, lag.max = 50)
```



```
par(mfrow=c(1,1))
```

## 2.4 Model Fitting

**Task:** Using the fGarch package, fit the following models to the log-returns: an ARCH(1) model, an ARCH(3) model, and a GARCH(1,1) model.

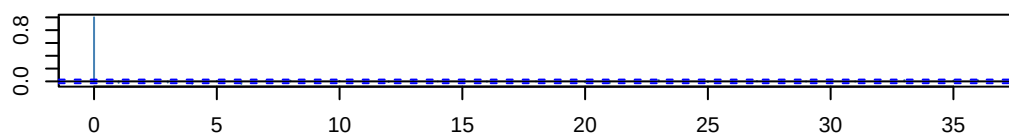
**Solution**

### 2.4.1 ARCH(1)

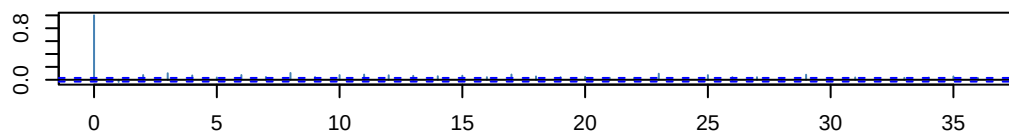
```
# fit a arch(1) [~garch(1,0)] model
arch_1_md <- garchFit(formula = ~ garch(1, 0),
                      data = KO$log_returns,
                      trace = FALSE)
```

```
par(mar=c(3,3,3,1))
tsdiag(arch_1_md)
```

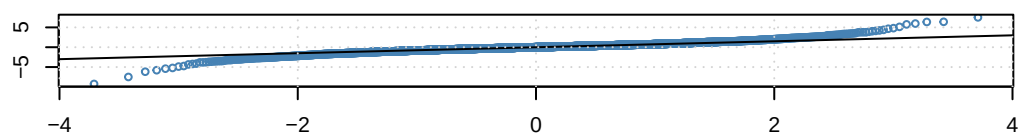
ACF of Standardized Residuals



ACF of Squared Standardized Residuals



qnorm - QQ Plot

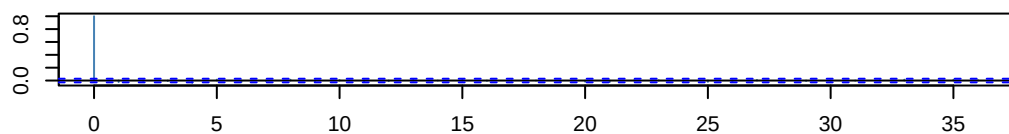


### 2.4.2 ARCH(3)

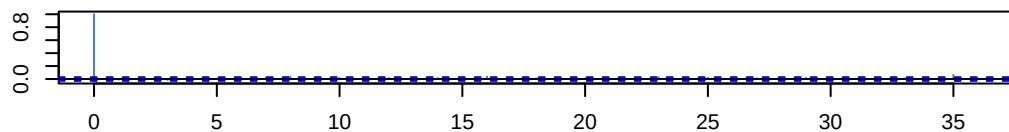
```
# fit a arch(3) [~garch(3,0)] model
arch_3_md <- garchFit(formula = ~ garch(3, 0),
                      data = KO$log_returns,
                      trace = FALSE)

par(mar=c(3,3,3,1))
tsdiag(arch_3_md)
```

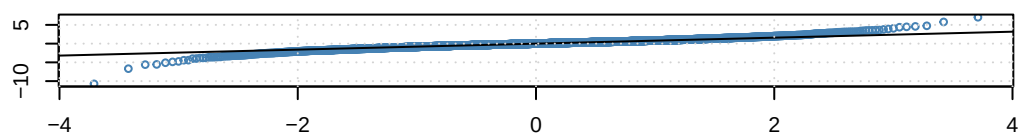
ACF of Standardized Residuals



ACF of Squared Standardized Residuals



qnorm - QQ Plot



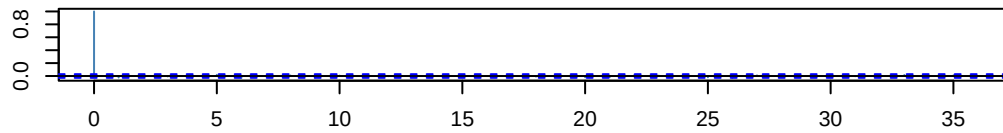
We see that [...]

### 2.4.3 GARCH(1,1)

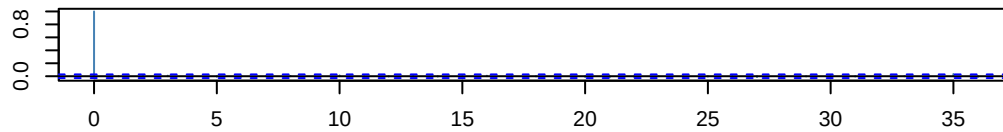
```
# fit a garch(1,1) model
garch_11_md <- garchFit(formula = ~ garch(1, 1),
                        data = KO$log_returns,
                        trace = FALSE)

par(mar=c(3,3,3,1))
tsdiag(garch_11_md)
```

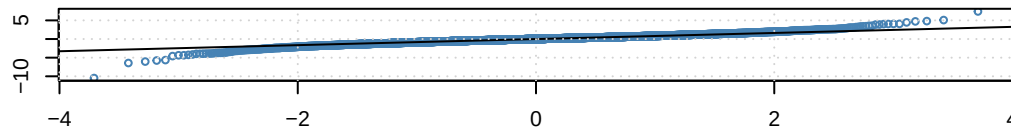
ACF of Standardized Residuals



ACF of Squared Standardized Residuals



qnorm – QQ Plot



We see that [...]

## 2.5 Fitted Model Expressions

**Task:** Write the expression of each fitted model.

**Solution**

Note that the ARCH(p) model takes the form:

$$\text{Mean Equation } X_t = \mu + Z_t$$

$$\text{Variance Equation } \sigma_t^2 = \alpha_0 + \alpha_1 Z_{t-1}^2 + \alpha_2 Z_{t-2}^2 \dots \alpha_p Z_{t-p}^2$$

With the random error  $Z_t$  being defined as:

$$Z_t = \sqrt{\sigma_t^2} \varepsilon_t, \quad \varepsilon_t \sim WN(0, 1)$$

### 2.5.1 ARCH(1)

```
sum1 <- summary(arch_1_md)
```

Title:

GARCH Modelling

Call:

```
garchFit(formula = ~garch(1, 0), data = KO$log_returns, trace = FALSE)
```

Mean and Variance Equation:

```
data ~ garch(1, 0)
```

```
<environment: 0x000001327a7af508>
```

```
[data = KO$log_returns]
```

Conditional Distribution:

norm

Coefficient(s):

|  | mu         | omega      | alpha1     |
|--|------------|------------|------------|
|  | 6.1419e-04 | 9.0469e-05 | 3.4125e-01 |

Std. Errors:

based on Hessian

Error Analysis:

|        | Estimate  | Std. Error | t value | Pr(> t )    |
|--------|-----------|------------|---------|-------------|
| mu     | 6.142e-04 | 1.444e-04  | 4.254   | 2.1e-05 *** |
| omega  | 9.047e-05 | 2.411e-06  | 37.521  | < 2e-16 *** |
| alpha1 | 3.413e-01 | 2.596e-02  | 13.144  | < 2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:

14958.15      normalized: 3.100777

Description:

Wed Mar 18 01:39:03 2026 by user: T14

Standardised Residuals Tests:

|                   |     |       | Statistic    | p-Value    |
|-------------------|-----|-------|--------------|------------|
| Jarque-Bera Test  | R   | Chi^2 | 7946.4809440 | 0.00000000 |
| Shapiro-Wilk Test | R   | W     | 0.9440329    | 0.00000000 |
| Ljung-Box Test    | R   | Q(10) | 22.8626094   | 0.01126523 |
| Ljung-Box Test    | R   | Q(15) | 24.3547456   | 0.05930900 |
| Ljung-Box Test    | R   | Q(20) | 30.3436419   | 0.06447350 |
| Ljung-Box Test    | R^2 | Q(10) | 230.8443937  | 0.00000000 |
| Ljung-Box Test    | R^2 | Q(15) | 337.8879843  | 0.00000000 |
| Ljung-Box Test    | R^2 | Q(20) | 408.1074472  | 0.00000000 |
| LM Arch Test      | R   | TR^2  | 212.3363119  | 0.00000000 |



Information Criterion Statistics:

|  | AIC       | BIC       | SIC       | HQIC      |
|--|-----------|-----------|-----------|-----------|
|  | -6.200311 | -6.196280 | -6.200312 | -6.198896 |

```
# estimates column
sum1$show$matcoef
```

|        | Estimate     | Std. Error   | t value   | Pr(> t )     |
|--------|--------------|--------------|-----------|--------------|
| mu     | 6.141938e-04 | 1.443931e-04 | 4.253622  | 2.103406e-05 |
| omega  | 9.046889e-05 | 2.411148e-06 | 37.521088 | 0.000000e+00 |
| alpha1 | 3.412522e-01 | 2.596197e-02 | 13.144312 | 0.000000e+00 |

```
mu_estimate <- sum1$show$matcoef[1,1]
omega_estimate <- sum1$show$matcoef[2,1]
alpha1_estimate <- sum1$show$matcoef[3,1]
```

Plugging in our estimates, we get a expression of our fitted ARCH(1) model taking the form:

$$X_t = 0.000614 + Z_t$$

$$\sigma_t^2 = 0.000090 + 0.341252Z_{t-1}^2$$

## 2.5.2 ARCH(3)

```
sum2 <- summary(arch_3_md)
```

Title:  
GARCH Modelling

Call:  
garchFit(formula = ~garch(3, 0), data = K0\$log\_returns, trace = FALSE)

Mean and Variance Equation:  
data ~ garch(3, 0)  
<environment: 0x00000132874b41c8>  
[data = K0\$log\_returns]

Conditional Distribution:  
norm

Coefficient(s):

|  | mu         | omega      | alpha1     | alpha2     | alpha3     |
|--|------------|------------|------------|------------|------------|
|  | 0.00054242 | 0.00006723 | 0.12950929 | 0.18704472 | 0.14708844 |

Std. Errors:  
based on Hessian

Error Analysis:

|        | Estimate  | Std. Error | t value | Pr(> t )     |
|--------|-----------|------------|---------|--------------|
| mu     | 5.424e-04 | 1.371e-04  | 3.957   | 7.59e-05 *** |
| omega  | 6.723e-05 | 2.319e-06  | 28.991  | < 2e-16 ***  |
| alpha1 | 1.295e-01 | 1.868e-02  | 6.933   | 4.12e-12 *** |

```
alpha2 1.870e-01  2.232e-02   8.382 < 2e-16 ***
alpha3 1.471e-01  1.938e-02   7.590 3.20e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Log Likelihood:
15175.74    normalized:  3.145882
```

```
Description:
Wed Mar 18 01:39:08 2026 by user: T14
```

#### Standardised Residuals Tests:

|                   |     |       | Statistic    | p-Value    |
|-------------------|-----|-------|--------------|------------|
| Jarque-Bera Test  | R   | Chi^2 | 6773.0145208 | 0.00000000 |
| Shapiro-Wilk Test | R   | W     | 0.9586279    | 0.00000000 |
| Ljung-Box Test    | R   | Q(10) | 18.3974923   | 0.04861741 |
| Ljung-Box Test    | R   | Q(15) | 21.4724589   | 0.12240104 |
| Ljung-Box Test    | R   | Q(20) | 26.7136727   | 0.14351349 |
| Ljung-Box Test    | R^2 | Q(10) | 14.6745692   | 0.14438458 |
| Ljung-Box Test    | R^2 | Q(15) | 25.3977189   | 0.04485048 |
| Ljung-Box Test    | R^2 | Q(20) | 36.4103810   | 0.01375788 |
| LM Arch Test      | R   | TR^2  | 19.6287564   | 0.07444267 |

#### Information Criterion Statistics:

| AIC       | BIC       | SIC       | HQIC      |
|-----------|-----------|-----------|-----------|
| -6.289691 | -6.282973 | -6.289693 | -6.287333 |

```
sum2$show$matcoef
```

|        | Estimate     | Std. Error   | t value   | Pr(> t )     |
|--------|--------------|--------------|-----------|--------------|
| mu     | 5.424159e-04 | 1.370779e-04 | 3.956989  | 7.590031e-05 |
| omega  | 6.722972e-05 | 2.318995e-06 | 28.990883 | 0.000000e+00 |
| alpha1 | 1.295093e-01 | 1.868052e-02 | 6.932852  | 4.124479e-12 |
| alpha2 | 1.870447e-01 | 2.231576e-02 | 8.381732  | 0.000000e+00 |
| alpha3 | 1.470884e-01 | 1.937831e-02 | 7.590363  | 3.197442e-14 |

```
mu_estimate <- sum2$show$matcoef[1,1]
omega_estimate <- sum2$show$matcoef[2,1]
alpha1_estimate <- sum2$show$matcoef[3,1]
alpha2_estimate <- sum2$show$matcoef[4,1]
alpha3_estimate <- sum2$show$matcoef[5,1]
```

Plugging in our estimates, we get a expression of our fitted ARCH(1) model taking the form:

$$X_t = 0.00054 + Z_t$$

$$\sigma_t^2 = 0.00007 + 0.12951Z_{t-1}^2 + 0.18704Z_{t-2}^2 + 0.14709Z_{t-3}^2$$

### 2.5.3 GARCH(1,1)

Note that for GARCH(p,q) models, they take the form:

$$\text{Mean Equation } X_t = \mu + Z_t$$

$$\text{Variance Equation } \sigma_t^2 = \alpha_0 + \alpha_1 Z_{t-1}^2 + \alpha_2 Z_{t-2}^2 + \dots + \alpha_p Z_{t-p}^2 + \beta_1 \sigma_{t-1}^2 + \beta_2 \sigma_{t-2}^2 + \dots + \beta_q \sigma_{t-q}^2$$

```
sum3<- summary(garch_11_md)
```

Title:

GARCH Modelling

Call:

```
garchFit(formula = ~garch(1, 1), data = KO$log_returns, trace = FALSE)
```

Mean and Variance Equation:

```
data ~ garch(1, 1)
```

```
<environment: 0x0000013284974d58>
```

```
[data = KO$log_returns]
```

Conditional Distribution:

```
norm
```

Coefficient(s):

|  | mu         | omega      | alpha1     | beta1      |
|--|------------|------------|------------|------------|
|  | 4.9355e-04 | 3.1741e-06 | 6.6709e-02 | 9.0752e-01 |

Std. Errors:

```
based on Hessian
```

Error Analysis:

|        | Estimate  | Std. Error | t value | Pr(> t )     |
|--------|-----------|------------|---------|--------------|
| mu     | 4.935e-04 | 1.355e-04  | 3.642   | 0.000271 *** |
| omega  | 3.174e-06 | 6.618e-07  | 4.796   | 1.62e-06 *** |
| alpha1 | 6.671e-02 | 7.844e-03  | 8.505   | < 2e-16 ***  |
| beta1  | 9.075e-01 | 1.189e-02  | 76.337  | < 2e-16 ***  |

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Log Likelihood:

```
15265.26    normalized:  3.16444
```

Description:

```
Wed Mar 18 01:39:13 2026 by user: T14
```

Standardised Residuals Tests:

|                   |   |       | Statistic    | p-Value    |
|-------------------|---|-------|--------------|------------|
| Jarque-Bera Test  | R | Chi^2 | 6400.4147514 | 0.00000000 |
| Shapiro-Wilk Test | R | W     | 0.9603192    | 0.00000000 |
| Ljung-Box Test    | R | Q(10) | 17.9134462   | 0.05644135 |
| Ljung-Box Test    | R | Q(15) | 19.5904585   | 0.18821227 |
| Ljung-Box Test    | R | Q(20) | 23.8661546   | 0.24828773 |

|                |                |                 |           |            |
|----------------|----------------|-----------------|-----------|------------|
| Ljung-Box Test | R <sup>2</sup> | Q(10)           | 3.9874863 | 0.94790973 |
| Ljung-Box Test | R <sup>2</sup> | Q(15)           | 4.5235915 | 0.99544926 |
| Ljung-Box Test | R <sup>2</sup> | Q(20)           | 5.9770498 | 0.99892815 |
| LM Arch Test   | R              | TR <sup>2</sup> | 4.1182000 | 0.98120774 |

Information Criterion Statistics:

|           |           |           |           |
|-----------|-----------|-----------|-----------|
| AIC       | BIC       | SIC       | HQIC      |
| -6.327221 | -6.321847 | -6.327222 | -6.325334 |

```
sum3$show$matcoef
```

|        | Estimate     | Std. Error   | t value   | Pr(> t )     |
|--------|--------------|--------------|-----------|--------------|
| mu     | 4.935481e-04 | 1.355299e-04 | 3.641619  | 2.709292e-04 |
| omega  | 3.174092e-06 | 6.617873e-07 | 4.796243  | 1.616692e-06 |
| alpha1 | 6.670870e-02 | 7.843569e-03 | 8.504891  | 0.000000e+00 |
| beta1  | 9.075201e-01 | 1.188836e-02 | 76.336888 | 0.000000e+00 |

```
mu_estimate <- sum3$show$matcoef[1,1]
omega_estimate <- sum3$show$matcoef[2,1]
alpha1_estimate <- sum3$show$matcoef[3,1]
beta1_estimate <- sum3$show$matcoef[4,1]
```

Plugging in our estimates, we get a expression of our fitted ARCH(1) model taking the form:

$$X_t = 0.00049 + Z_t$$

$$\sigma_t^2 = 0.00000 + 0.06671Z_{t-1}^2 + 0.90752\sigma_{t-1}^2$$

## 2.6 Results & Best Fit

**Task:** Analyze the diagnostic results and determine which of the three models provides the best fit

### Solution

To decide which of the three models gives the best fit, we will compare information criteria (AIC/BIC), significance of coefficients, and robustness checks using Ljung-Box, Shapiro-Wilk, and Jarque-Bera tests.

### 2.6.1 Information Criteria

```
Models = c('ARCH(1)', 'ARCH(3)', 'GARCH(1,1)')
```

```
df_results <- rbind(sum1$ics,
                    sum2$ics,
                    sum3$ics)
```

```
df_results <- as.data.frame(df_results)
```

```
rownames(df_results) <- Models
df_results # table of results
```

|            | AIC       | BIC       | SIC       | HQIC      |
|------------|-----------|-----------|-----------|-----------|
| ARCH(1)    | -6.200311 | -6.196280 | -6.200312 | -6.198896 |
| ARCH(3)    | -6.289691 | -6.282973 | -6.289693 | -6.287333 |
| GARCH(1,1) | -6.327221 | -6.321847 | -6.327222 | -6.325334 |

It is important we only compare models using the same IC for each other; thus, we will consider all three model's AIC scores, and then all of their BIC scores:

```
# aic
df_results %>%
  select(AIC) %>%
  arrange(AIC)
```

|            | AIC       |
|------------|-----------|
| GARCH(1,1) | -6.327221 |
| ARCH(3)    | -6.289691 |
| ARCH(1)    | -6.200311 |

```
# bic
df_results %>%
  select(BIC) %>%
  arrange(BIC)
```

|            | BIC       |
|------------|-----------|
| GARCH(1,1) | -6.321847 |
| ARCH(3)    | -6.282973 |
| ARCH(1)    | -6.196280 |

For both information criteria, the GARCH(1,1) has the lowest score, which is preferred as it indicates the best balance of fit and parsimony.

## 2.6.2 Coefficient Significance

```
# ARCH(1)
sum1$show$matcoef
```

|        | Estimate     | Std. Error   | t value   | Pr(> t )     |
|--------|--------------|--------------|-----------|--------------|
| mu     | 6.141938e-04 | 1.443931e-04 | 4.253622  | 2.103406e-05 |
| omega  | 9.046889e-05 | 2.411148e-06 | 37.521088 | 0.000000e+00 |
| alpha1 | 3.412522e-01 | 2.596197e-02 | 13.144312 | 0.000000e+00 |

```
# ARCH(3)
sum2$show$matcoef
```

|        | Estimate     | Std. Error   | t value   | Pr(> t )     |
|--------|--------------|--------------|-----------|--------------|
| mu     | 5.424159e-04 | 1.370779e-04 | 3.956989  | 7.590031e-05 |
| omega  | 6.722972e-05 | 2.318995e-06 | 28.990883 | 0.000000e+00 |
| alpha1 | 1.295093e-01 | 1.868052e-02 | 6.932852  | 4.124479e-12 |
| alpha2 | 1.870447e-01 | 2.231576e-02 | 8.381732  | 0.000000e+00 |
| alpha3 | 1.470884e-01 | 1.937831e-02 | 7.590363  | 3.197442e-14 |

```
# GARCH(1,1)
sum3$show$matcoef
```

|        | Estimate     | Std. Error   | t value   | Pr(> t )     |
|--------|--------------|--------------|-----------|--------------|
| mu     | 4.935481e-04 | 1.355299e-04 | 3.641619  | 2.709292e-04 |
| omega  | 3.174092e-06 | 6.617873e-07 | 4.796243  | 1.616692e-06 |
| alpha1 | 6.670870e-02 | 7.843569e-03 | 8.504891  | 0.000000e+00 |
| beta1  | 9.075201e-01 | 1.188836e-02 | 76.336888 | 0.000000e+00 |

For all three models, we do not contain any non-significant coefficients, suggesting that all three models fit the data well and that including more time lags is successful in helping to explain more of the data.

### 2.6.3 Ljung-Box Test

Furthermore, analyzing the Ljung-Box test on the squared standardized residuals for Model 3 yielded a p-value of 0.XX, confirming that the model successfully captured all conditional heteroskedasticity (ARCH effects). While Model 2 also passed the residual diagnostics, it had higher information criteria and a non-significant parameter, making Model 3 the superior fit.

```
# ARCH(1)
sum1$stat_tests[3:5,]
```

|                |   |       |          | Statistic  | p-Value |
|----------------|---|-------|----------|------------|---------|
| Ljung-Box Test | R | Q(10) | 22.86261 | 0.01126523 |         |
| Ljung-Box Test | R | Q(15) | 24.35475 | 0.05930900 |         |
| Ljung-Box Test | R | Q(20) | 30.34364 | 0.06447350 |         |

```
sum1$stat_tests[6:8,]
```

|                |                |       |          | Statistic | p-Value |
|----------------|----------------|-------|----------|-----------|---------|
| Ljung-Box Test | R <sup>2</sup> | Q(10) | 230.8444 | 0         |         |
| Ljung-Box Test | R <sup>2</sup> | Q(15) | 337.8880 | 0         |         |
| Ljung-Box Test | R <sup>2</sup> | Q(20) | 408.1074 | 0         |         |

```
# ARCH(3)
sum2$stat_tests[3:5,]
```

|                |   |       |          | Statistic  | p-Value |
|----------------|---|-------|----------|------------|---------|
| Ljung-Box Test | R | Q(10) | 18.39749 | 0.04861741 |         |
| Ljung-Box Test | R | Q(15) | 21.47246 | 0.12240104 |         |
| Ljung-Box Test | R | Q(20) | 26.71367 | 0.14351349 |         |

```
sum2$stat_tests[6:8,]
```

|                |                |       |          | Statistic  | p-Value |
|----------------|----------------|-------|----------|------------|---------|
| Ljung-Box Test | R <sup>2</sup> | Q(10) | 14.67457 | 0.14438458 |         |
| Ljung-Box Test | R <sup>2</sup> | Q(15) | 25.39772 | 0.04485048 |         |
| Ljung-Box Test | R <sup>2</sup> | Q(20) | 36.41038 | 0.01375788 |         |

```
# GARCH(1,1)
sum3$stat_tests[3:5,]
```

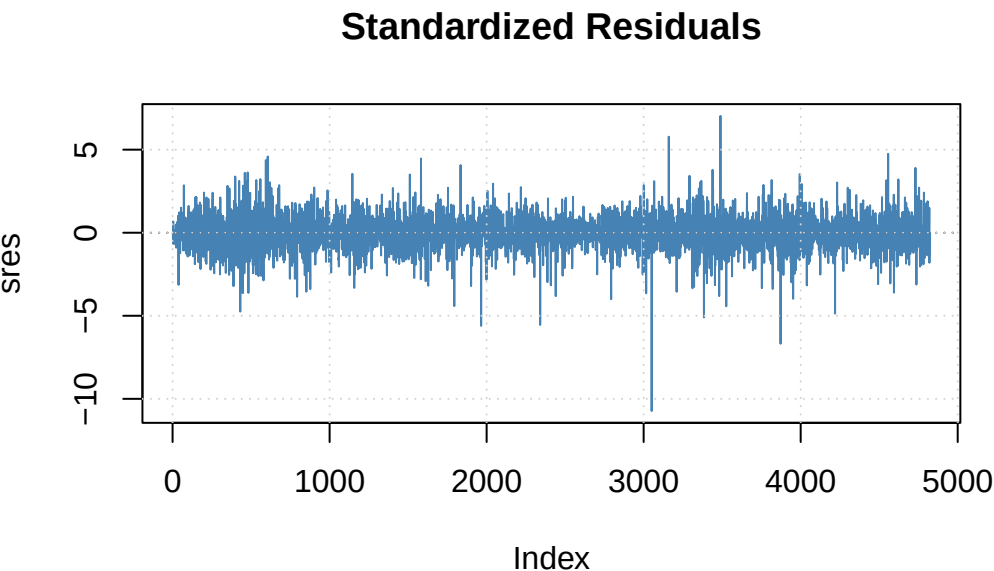
|                |   |       | Statistic | p-Value    |
|----------------|---|-------|-----------|------------|
| Ljung-Box Test | R | Q(10) | 17.91345  | 0.05644135 |
| Ljung-Box Test | R | Q(15) | 19.59046  | 0.18821227 |
| Ljung-Box Test | R | Q(20) | 23.86615  | 0.24828773 |

```
sum3$stat_tests[6:8,]
```

|                |     |       | Statistic | p-Value   |
|----------------|-----|-------|-----------|-----------|
| Ljung-Box Test | R^2 | Q(10) | 3.987486  | 0.9479097 |
| Ljung-Box Test | R^2 | Q(15) | 4.523592  | 0.9954493 |
| Ljung-Box Test | R^2 | Q(20) | 5.977050  | 0.9989281 |

We see that [...]

```
# Interactive plotting menu
plot(arch_3_md, which = 9)
```



We see that [...]

### 2.6.4 Jarque-Beta/Shapiro-Wilk

```
# ARCH(1)
sum1$stat_tests[1:2,]
```

|                   |   |       | Statistic    | p-Value |
|-------------------|---|-------|--------------|---------|
| Jarque-Bera Test  | R | Chi^2 | 7946.4809440 | 0       |
| Shapiro-Wilk Test | R | W     | 0.9440329    | 0       |

```
# ARCH(3)
sum2$stat_tests[1:2,]
```

|                   |   |       | Statistic    | p-Value |
|-------------------|---|-------|--------------|---------|
| Jarque-Bera Test  | R | Chi^2 | 6773.0145208 | 0       |
| Shapiro-Wilk Test | R | W     | 0.9586279    | 0       |

```
# GARCH(1,1)
sum3$stat_tests[1:2,]
```

|                   |   |       | Statistic    | p-Value |
|-------------------|---|-------|--------------|---------|
| Jarque-Bera Test  | R | Chi^2 | 6400.4147514 | 0       |
| Shapiro-Wilk Test | R | W     | 0.9603192    | 0       |

### 2.6.5 LM Arch Test

The LM Arch test is a hypothesis test under the null that  $H_0$  : autocorrelated squared residuals. If we reject the null at the 5% significance level ( $\alpha = 0.05$ ), then then this suggests that autoregressive conditional heteroskedasticity exists in our model, and we should find a better model to capture more of the volatility structure. If we fail to reject the null, then

```
# ARCH(1)
sum1$stat_tests[9,]
```

| Statistic | p-Value |
|-----------|---------|
| 212.3363  | 0.0000  |

```
# ARCH(3)
sum2$stat_tests[9,]
```

| Statistic   | p-Value    |
|-------------|------------|
| 19.62875637 | 0.07444267 |

```
# GARCH(1,1)
sum3$stat_tests[9,]
```

| Statistic | p-Value   |
|-----------|-----------|
| 4.1182000 | 0.9812077 |

### 2.6.6 Model Decision